Transportation Problem using Python (10 Problems with Different Data)

Problem 1

Aim:

To solve the transportation problem using Python and minimize the total transportation cost.

Python Code:

```
import numpy as np
from scipy.optimize import linprog

cost = [2, 4, 5, 3]
A_eq = [[1, 1, 0, 0], [0, 0, 1, 1]]
b_eq = [20, 30]

result = linprog(cost, A_eq=A_eq, b_eq=b_eq)
print('Minimum Transportation Cost =', result.fun)
```

Output:

```
Minimum Transportation Cost = 130.0
```

Problem 2

Aim:

To solve the transportation problem using Python and minimize the total transportation cost.

Python Code:

```
import numpy as np
from scipy.optimize import linprog

cost = [6, 3, 2, 4]
A_eq = [[1, 0, 1, 0], [0, 1, 0, 1]]
b_eq = [25, 35]

result = linprog(cost, A_eq=A_eq, b_eq=b_eq)
print('Minimum Transportation Cost =', result.fun)
```

Output:

```
Minimum Transportation Cost = 155.0
```

Problem 3

Aim:

To solve the transportation problem using Python and minimize the total transportation cost.

Python Code:

```
import numpy as np
from scipy.optimize import linprog

cost = [5, 7, 4, 6]
A_eq = [[1, 1, 0, 0], [0, 0, 1, 1]]
b_eq = [40, 20]

result = linprog(cost, A_eq=A_eq, b_eq=b_eq)
print('Minimum Transportation Cost =', result.fun)
```

Output:

```
Minimum Transportation Cost = 280.0
```

Problem 4

Aim:

To solve the transportation problem using Python and minimize the total transportation cost.

Python Code:

```
import numpy as np
from scipy.optimize import linprog

cost = [3, 8, 6, 5]
A_eq = [[1, 0, 1, 0], [0, 1, 0, 1]]
b_eq = [15, 45]

result = linprog(cost, A_eq=A_eq, b_eq=b_eq)
print('Minimum Transportation Cost =', result.fun)
```

Output:

```
Minimum Transportation Cost = 270.0
```

Problem 5

Aim:

To solve the transportation problem using Python and minimize the total transportation

Python Code:

```
import numpy as np
from scipy.optimize import linprog

cost = [9, 4, 3, 7]
A_eq = [[1, 1, 0, 0], [0, 0, 1, 1]]
b_eq = [50, 30]

result = linprog(cost, A_eq=A_eq, b_eq=b_eq)
print('Minimum Transportation Cost =', result.fun)
```

Output:

```
Minimum Transportation Cost = 290.0
```

Problem 6

Aim:

To solve the transportation problem using Python and minimize the total transportation cost.

Python Code:

```
import numpy as np
from scipy.optimize import linprog

cost = [4, 6, 8, 5]
A_eq = [[1, 0, 1, 0], [0, 1, 0, 1]]
b_eq = [60, 20]

result = linprog(cost, A_eq=A_eq, b_eq=b_eq)
print('Minimum Transportation Cost =', result.fun)
```

Output:

```
Minimum Transportation Cost = 340.0
```

Problem 7

Aim:

To solve the transportation problem using Python and minimize the total transportation cost.

Python Code:

```
import numpy as np
from scipy.optimize import linprog

cost = [7, 5, 6, 4]
A_eq = [[1, 1, 0, 0], [0, 0, 1, 1]]
b_eq = [35, 25]

result = linprog(cost, A_eq=A_eq, b_eq=b_eq)
print('Minimum Transportation Cost =', result.fun)
```

Output:

```
Minimum Transportation Cost = 275.0
```

Problem 8

Aim:

To solve the transportation problem using Python and minimize the total transportation cost.

Python Code:

```
import numpy as np
from scipy.optimize import linprog

cost = [8, 6, 4, 3]
A_eq = [[1, 0, 1, 0], [0, 1, 0, 1]]
b_eq = [45, 15]

result = linprog(cost, A_eq=A_eq, b_eq=b_eq)
print('Minimum Transportation Cost =', result.fun)
```

Output:

```
Minimum Transportation Cost = 225.0
```

Problem 9

Aim:

To solve the transportation problem using Python and minimize the total transportation cost.

Python Code:

```
import numpy as np
from scipy.optimize import linprog

cost = [6, 9, 5, 7]
A_eq = [[1, 1, 0, 0], [0, 0, 1, 1]]
b_eq = [55, 35]

result = linprog(cost, A_eq=A_eq, b_eq=b_eq)
print('Minimum Transportation Cost =', result.fun)
```

Output:

```
Minimum Transportation Cost = 505.0
```

Problem 10

Aim:

To solve the transportation problem using Python and minimize the total transportation cost.

Python Code:

```
import numpy as np
from scipy.optimize import linprog

cost = [3, 5, 4, 6]
A_eq = [[1, 0, 1, 0], [0, 1, 0, 1]]
b_eq = [30, 40]

result = linprog(cost, A_eq=A_eq, b_eq=b_eq)
print('Minimum Transportation Cost =', result.fun)
```

Output:

```
Minimum Transportation Cost = 290.0
```